

LITTLE FLATLAND ISLAND, ON, Canada

WMO: 714680

Lat: 49.690N

Lon: 88.306W

Elev: 261

StdP: 98.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-26.9	-32.2	0.2	-29.2	-29.6	0.3	-26.8	14.4	-12.3	13.2	-13.7	3.7	290	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	25.0	19.7	23.5	18.9	22.1	18.0	20.6	23.8	19.6	22.4	18.6	21.2	2.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.6	22.4	18.4	13.7	21.3	17.5	13.0	20.3	60.6	23.8	57.2	22.6	54.0	21.2	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
13.5	12.0	10.6	DB	-32.1	28.7	3.6	1.7	-34.7	29.9	-36.8	30.9	-38.8	31.9	-41.4	33.1	
			WB	-32.2	22.3	3.3	1.3	-34.6	23.2	-36.6	24.0	-38.5	24.7	-40.9	25.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-14.5	-14.0	-7.4	-0.4	6.3	12.7	17.6	17.8	13.4	6.4	-0.2	-5.5
	DBStd	12.04	7.41	7.12	6.86	4.78	3.65	3.50	2.91	2.78	3.66	3.93	4.80	5.14
	HDD10.0	3346	759	672	539	311	125	14	0	0	12	124	307	482
	HDD18.3	5738	1018	906	798	561	373	172	49	43	152	369	557	740
	CDD10.0	709	0	0	0	0	11	94	236	242	113	14	0	0
	CDD18.3	58	0	0	0	0	0	2	26	27	3	0	0	0
	CDH23.3	143	0	0	0	0	1	8	73	59	2	0	0	0
	CDH26.7	10	0	0	0	0	0	1	6	3	0	0	0	0
Wind	WSAvg	4.6	4.8	4.3	4.1	4.0	3.7	3.4	3.8	4.3	4.9	5.9	6.1	6.2
Precipitation	PrecAvg	879	47	34	42	67	88	100	96	89	104	91	68	53
	PrecMax	1050	112	84	83	196	207	204	170	137	215	167	133	95
	PrecMin	646	11	11	11	18	48	46	57	42	43	34	25	26
	PrecStd	101	19	17	16	33	31	35	29	26	37	34	26	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.2	7.7	12.4	20.0	24.3	27.4	27.0	23.2	17.2	11.3	5.3
		MCWB	0.9	1.7	5.1	6.7	12.6	17.2	21.0	20.7	18.2	13.5	8.7	3.9
	2%	DB	0.5	1.5	4.7	8.9	16.5	21.9	25.2	25.0	21.4	15.1	9.2	3.7
		MCWB	0.1	0.7	2.6	4.8	10.5	16.6	19.8	20.1	17.6	12.4	7.7	2.7
	5%	DB	-0.8	0.0	3.1	6.9	13.7	20.1	23.7	23.5	19.7	13.4	7.7	2.5
		MCWB	-1.3	-0.7	1.7	3.6	9.5	15.6	19.2	19.1	16.7	11.4	6.2	1.6
	10%	DB	-3.2	-2.7	1.7	5.1	11.6	18.1	22.2	22.0	18.2	12.0	6.3	1.3
		MCWB	-3.6	-3.2	0.9	2.7	8.3	14.6	18.3	18.3	16.0	10.2	5.1	0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	2.0	5.9	7.2	14.0	18.7	22.2	22.2	19.4	14.4	9.3	4.1
		MCDB	1.5	2.7	6.8	11.0	17.5	22.3	25.6	25.4	21.7	16.5	10.5	4.9
	2%	WB	0.2	0.8	2.9	5.2	11.5	17.2	20.7	20.7	18.1	13.0	7.7	2.9
		MCDB	0.5	1.4	4.3	8.0	14.7	20.7	24.1	23.7	20.2	14.4	9.0	3.6
	5%	WB	-1.3	-0.7	1.8	3.9	9.9	16.1	19.7	19.8	17.3	11.7	6.5	1.8
		MCDB	-0.9	-0.1	2.8	6.4	12.9	19.2	22.7	22.5	19.2	13.2	7.5	2.3
	10%	WB	-3.7	-3.2	0.9	2.8	8.4	14.9	18.7	18.9	16.3	10.3	5.1	0.7
		MCDB	-3.3	-2.7	1.8	4.9	11.1	17.5	21.4	21.4	18.0	11.8	6.3	1.3
Mean Daily Temperature Range		MDBR	7.7	9.8	9.7	8.3	8.3	8.2	7.7	6.9	6.2	4.5	4.4	4.9
	5% DB	MCDBR	7.2	8.5	8.9	10.0	12.3	11.0	9.7	8.9	7.8	6.7	5.2	4.8
		MCWBR	7.0	7.8	6.9	6.3	7.5	6.1	5.3	5.0	4.7	4.7	4.2	4.4
		MCDBR	7.3	8.5	7.8	9.2	11.1	9.6	9.1	8.3	7.0	6.3	5.2	4.8
	5% WB	MCWBR	7.1	7.9	6.2	6.1	7.3	5.9	5.4	5.0	4.5	4.6	4.3	4.6
Clear-Sky Solar Irradiance	taub	0.246	0.266	0.299	0.343	0.353	0.370	0.390	0.367	0.350	0.308	0.283	0.259	
	taud	2.307	2.266	2.305	2.372	2.429	2.415	2.388	2.448	2.492	2.575	2.429	2.279	
	Ebn at Noon	819	885	911	904	905	889	863	869	850	838	768	736	
	Edh at Noon	67	90	105	110	109	112	113	102	88	68	60	60	
All-Sky Solar Radiation	RadAvg	1.24	2.42	3.74	4.82	5.24	5.63	5.61	4.75	3.38	1.96	1.00	0.83	
	RadStd	0.15	0.21	0.32	0.45	0.29	0.43	0.39	0.39	0.27	0.29	0.12	0.12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page